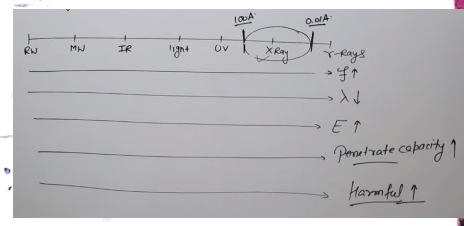
X-Rays

→ X-rays are highly energetic EM-waves, having large f, small λ and high penetration capacity.

→ EM radiations having wavelength in the range of 0.01Å to 100Å in air/vacuum are termed as X-rays.



→ Two types of X-Ray:

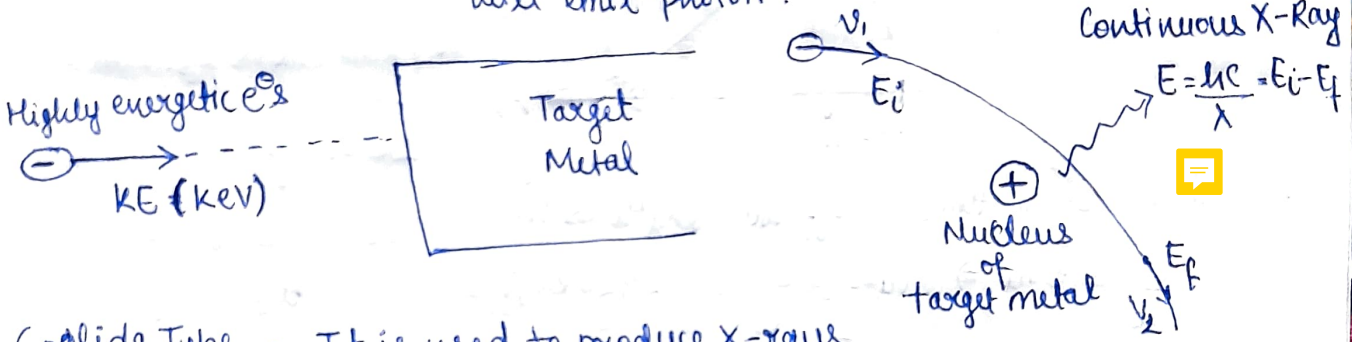
① Soft X-Ray:
- $\lambda > 10Å$
- low energy
- low penetration

② Hard X-Ray:
- $\lambda < 1Å$
- Highly energetic
- Highly penetrative

used for security purposes

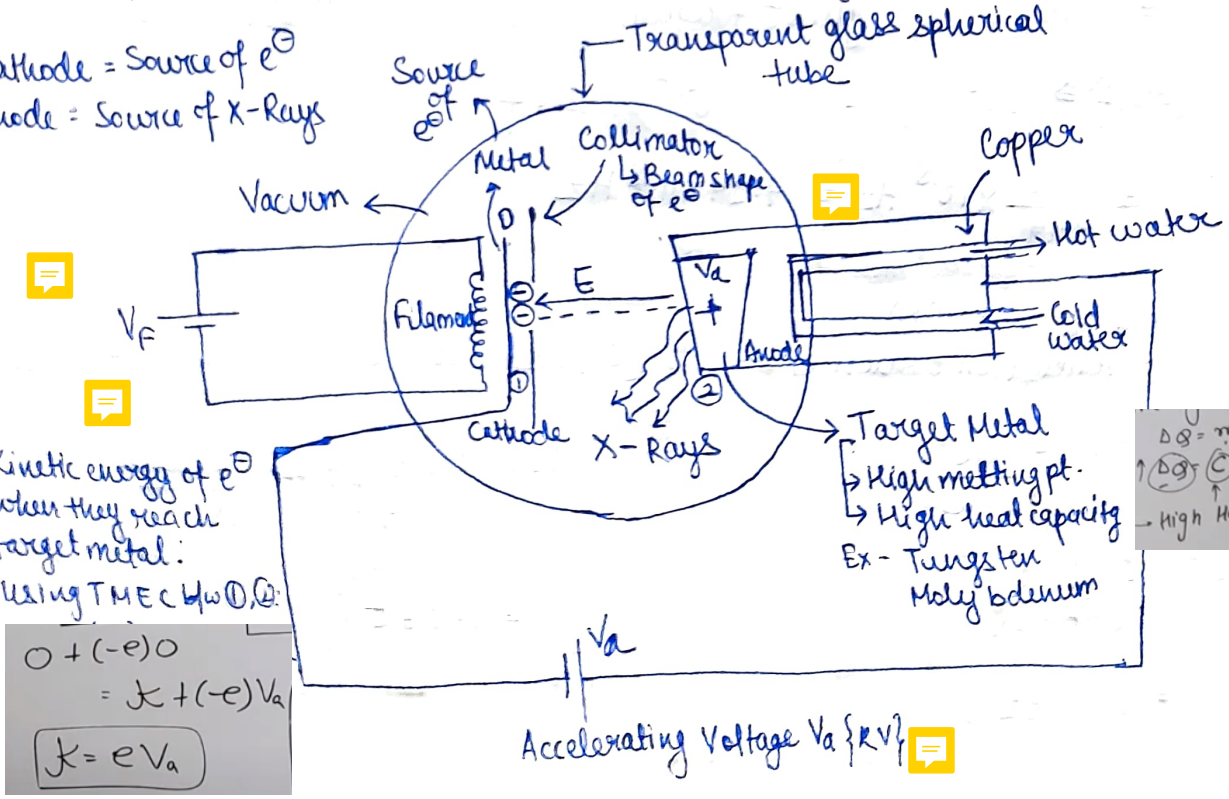
Bremsstrahlung effect: It states if highly energetic $e^{\ominus}$ are bombarded on a target metal, then nucleus of target metal retards the motion of $e^{\ominus}$ and any loss in KE of $e^{\ominus}$ will emit photon.

to stop/ to brake — radiation



Coolidg Tube : It is used to produce X-rays
X-Ray Tube : It is based on Bremsstrahlung effect.

Cathode = Source of $e^{\ominus}$
Anode = Source of X-Rays



Kinetic energy of $e^{\ominus}$ when they reach target metal:
Using TMEC b/w ①, ②:

$$0 + (-e)0 = K + (-e)V_a$$

$$K = eV_a$$

$\Delta Q = ms\Delta T$

$\frac{\Delta Q}{\Delta T} = C$

↑ High Heat Capacity

Cutoff Wavelength (λ_c): Smallest wavelength of X-Ray spectrum. So the photon must be most energetic.

$$\frac{hc}{\lambda_c} = eV_a \Rightarrow \lambda_c = \frac{hc}{eV_A} = \frac{12420\ eV\text{-}\mathring{A}}{V_a\ \{\text{Volts}\}}$$

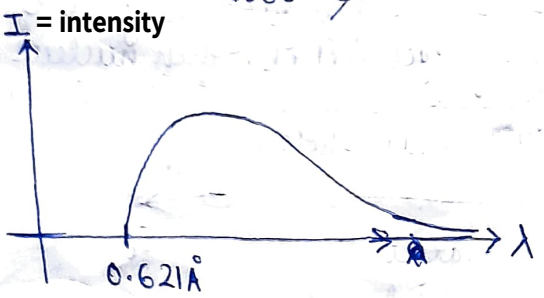
$$\therefore \boxed{\lambda_c = \frac{12420\ \mathring{A}}{V_a(\text{Volts})}}$$

Q Find cutoff wavelength if accelerating voltage is 20KV.

$V_a = 20{,}000V$

$K = eV_a = 20000\ eV$

$$K = \frac{hc}{\lambda_c} \Rightarrow \lambda_c = \frac{hc}{K} = \frac{12420\ eV\mathring{A}}{20000\ eV} = 0.621\mathring{A}$$



Q Find V_a such that cutoff wavelength is 1Å.

$$\frac{hc}{\lambda_c} = eV_a \Rightarrow V_a = \frac{hc}{\lambda_c e} = \frac{12420\ eV\mathring{A}}{1\mathring{A}\ e} = 12420\ \text{Volts}$$

Q If $V_a = 30KV$, Find highest frequency of X-Ray emitted.

$$eV_a = hf \Rightarrow 30000\ eV = 4.14 \times 10^{-15}\ eV\text{-sec}\ f$$

$$f = \frac{30000}{4.14} \times 10^{15}\ Hz = 7.24 \times 10^{18}\ Hz$$

Q ① Find λ_c if $V_a = 40KV$

② Find λ of radiation which emits if $e^{\ominus}$ looses 30% of its energy during collision.

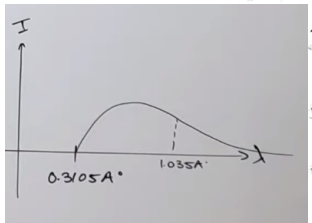
① $40000\ eV = \frac{hc}{\lambda_c}$

$$\lambda_c = \frac{12420\ eV\mathring{A}}{40000\ eV}$$

$$\lambda_c = 0.3105\mathring{A}$$

② $40000\ eV \times \frac{30}{100} = \frac{12420\ eV\mathring{A}}{\lambda}$

$$\boxed{\lambda = 1.035\mathring{A}}$$



Note:

① $V_a \uparrow \Rightarrow$ Energy of $e^{\ominus} \uparrow \Rightarrow$ Energy lost $\uparrow \rightarrow$ Photons Energy $\uparrow \Rightarrow \lambda \downarrow \Rightarrow$ Hard X-Rays

$V_a \downarrow \Rightarrow$ Energy of $e^{\ominus} \downarrow \Rightarrow$ Energy lost $\downarrow \Rightarrow$ Photons Energy $\downarrow \Rightarrow \lambda \uparrow \Rightarrow$ Soft X-Rays.

② $\boxed{\lambda_c \propto \frac{1}{V_a}}$

③ Potential across filament $(V_F) \uparrow \Rightarrow$ Heat $\uparrow \Rightarrow$ Metal $e^{\ominus}$ ejection $\uparrow \Rightarrow$ no. of photon $\uparrow \Rightarrow$ X-ray Intensity $\uparrow$.

∴ To change Quality of X-ray, we need to change λ_c for which we change V_a.
To change Intensity of X-ray, we change V_F.

Q Given $V_a = 15KV$. Find KE and speed of $e^{\ominus}$ when it reaches anode. Also find λ_c.

$KE = eV_a$
$= 15000 \times 1.6 \times 10^{-19}$
$= 24 \times 10^{-16} J$

$KE = \frac{1}{2}mv^2 = 24 \times 10^{-16} J$

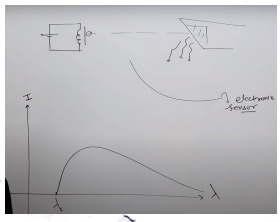
$v^2 = \frac{2 \times 24 \times 10^{-16}}{9.1 \times 10^{-31}}$

$v^2 = 52.74 \times 10^{14}$

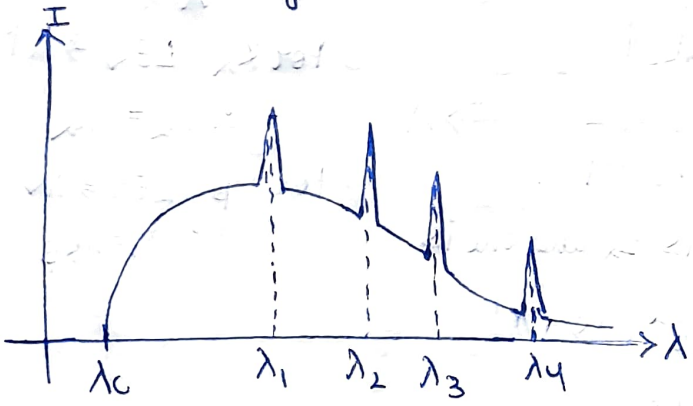
$\boxed{v = 7.26 \times 10^7 m/s}$

$\lambda_c = \frac{hc}{eV_a} = \frac{12420 \text{ eV Å}}{15000 \text{ eV}}$

$\boxed{\lambda_c = 0.828 Å}$

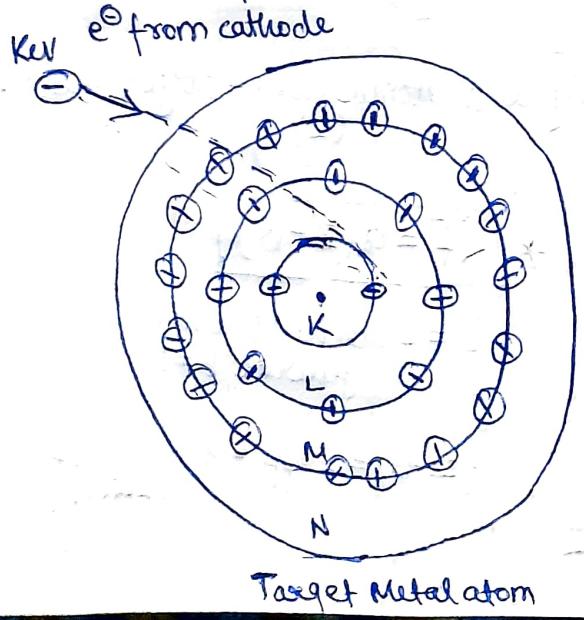


Characteristic X-Rays:

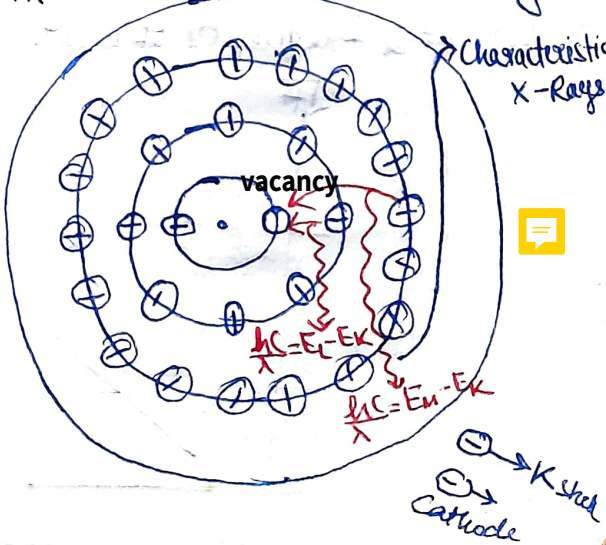


When we plot the intensity v/s λ graph of the photons emitted from target metal we observe that there are sudden peaks in graph. That means for some values of λ, apart from continuous X-Rays there must be some more X-Rays. These are characteristic X-Rays.

Generation of characteristic X-Rays:



If $e^{\ominus}$ from cathode comes and knocks of $e^{\ominus}$ in K-shell and creates vacancy.

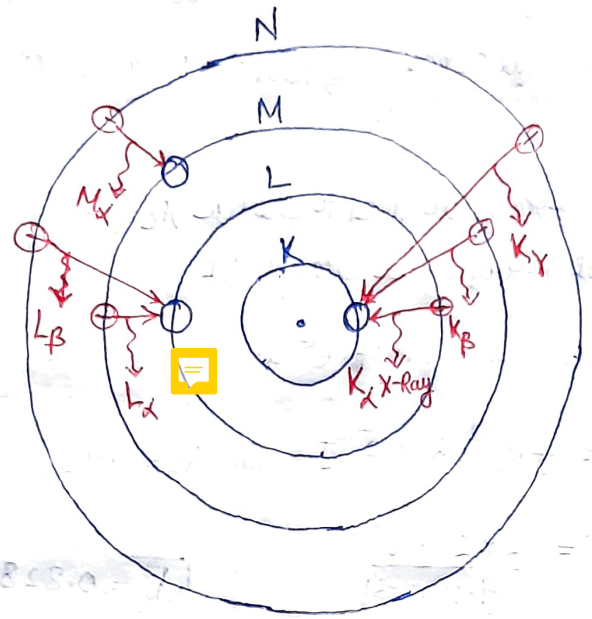


Spectral Series of Characteristic X-Ray:

$\frac{hc}{\lambda_{L\alpha}} = E_M - E_L$

$\frac{hc}{\lambda_{L\beta}} = E_N - E_L$

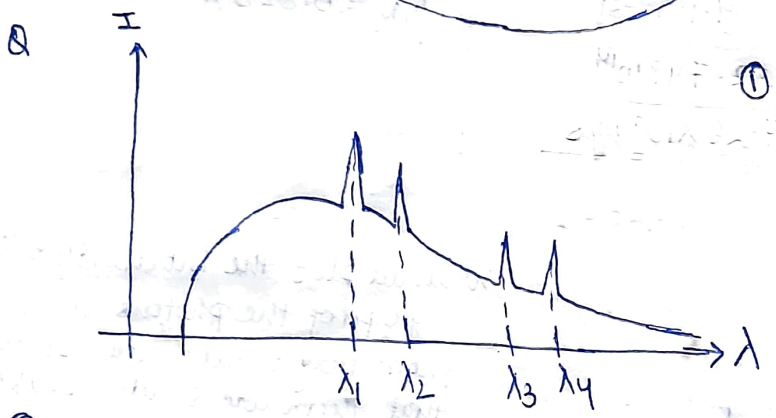
$\frac{hc}{\lambda_{M\alpha}} = E_N - E_M$



$\frac{hc}{\lambda_{K\alpha}} = E_L - E_K$

$\frac{hc}{\lambda_{K\beta}} = E_M - E_K$

$\frac{hc}{\lambda_{K\gamma}} = E_N - E_K$



① Given out of λ_1 and λ_2 one is K_α and one is K_β. Find which one.

$\Delta E \propto \frac{1}{\lambda}$

For K_α, $\Delta E \downarrow \Rightarrow \lambda \uparrow$

$\therefore \lambda_2 = \lambda_{K\alpha}$

For K_β, $\Delta E \uparrow \Rightarrow \lambda \downarrow$

$\therefore \lambda_1 = \lambda_{K\beta}$

② Given out of λ_3 and λ_4 one is L_α and one is L_β. Find which one.

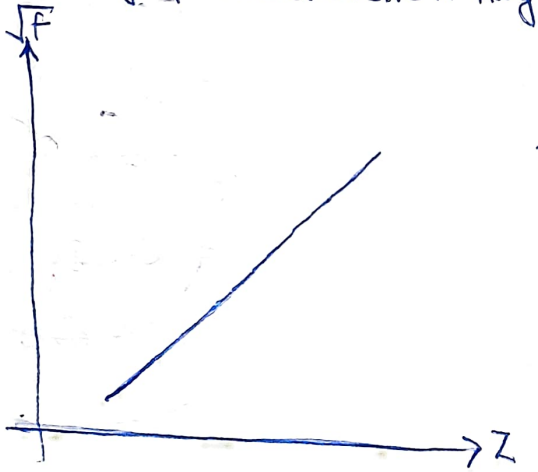
For L_α, $\Delta E \downarrow \Rightarrow \lambda \uparrow$ $\quad\quad$ For L_β, $\Delta E \uparrow, \lambda \downarrow$

$\lambda_4 = \lambda_{L\alpha}$ $\quad\quad$ $\lambda_3 = \lambda_{L\beta}$

Moseley's law:

→ It is an experimental law.

→ He related frequency of characteristic X-Ray and atomic no. of target metal. (Z)

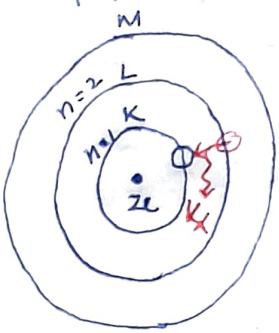


* $\boxed{\sqrt{f} = a(Z-b)}$

b = Screening const.
in general b = 1

$(Z-b) = Z_{eff}$ = effective atomic no.

Proof for H-like atom:



$$\frac{hc}{\lambda_{K_\alpha}} = E_L - E_K$$

$$\frac{hc}{\lambda_{K_\alpha}} = \frac{-13.6\,ev\,(z-b)^2}{2^2} + \frac{13.6\,ev\,(z-b)^2}{1^2}$$

$$\frac{1}{\lambda_{K_\alpha}} = \frac{13.6\,ev}{hc}(z-b)^2\left\{\frac{1}{1^2} - \frac{1}{2^2}\right\}$$

$$\frac{1}{\lambda_{K_\alpha}} = R(z-b)^2 \times \frac{3}{4}$$

$$f_{K_\alpha} = \frac{c}{\lambda_{K_\alpha}} = \frac{3Rc}{4}(z-b)^2 \Rightarrow \boxed{\sqrt{f_{K_\alpha}} = \sqrt{\frac{3Rc}{4}}(z-b)} \equiv \sqrt{f} = a(z-b)$$

Similarly we can prove for K_β, K_γ etc.

$a = \sqrt{\frac{3Rc}{4}}$ for n=2 to n=1 transition

Q For zinc ($z=30$), K_α X-Ray have $\lambda = 1.42\text{Å}$, then what will be λ of K_α X-ray if we replace the ~~target~~ metal of Molybdenum ($z=42$)?

$$\sqrt{f} = a(z-b)$$

$$= \sqrt{\frac{c}{\lambda}} = a(z-b)$$

$$= \sqrt{\frac{\lambda_{Mo}}{1.42\text{Å}}} = \frac{29}{41}$$

For Zn $\Rightarrow$ $\sqrt{\frac{c}{\lambda_{Zn}}} = a(30-1)$ — ①

For Mo $\Rightarrow$ $\sqrt{\frac{c}{\lambda_{Mo}}} = a(42-1)$ — ②

$$\lambda_{Mo} = \left(\frac{29}{41}\right)^2 \times 1.42\text{Å} = 5 \times 1.42\text{Å}$$

$$\lambda_{Mo} = 7.1\text{Å}$$

Q Frequency of K_α X-Ray, if target metal is La ($z=57$) is 1.9×10^{18} Hz. Find f of K_α X-Ray if target metal is copper (Cu) {$z=29$}.

$$\sqrt{f} = a(z-b)$$

$$\sqrt{f_{La}} = a(57-1)$$

$$\sqrt{f_{Cu}} = a(29-1)$$

$$\Rightarrow f_{Cu} = \frac{1.9 \times 10^{18}}{4}\ \text{Hz}$$

$$\sqrt{\frac{f_{Cu}}{1.9 \times 10^{18}}} = \frac{28}{56}$$

Q For a target metal, $K_\alpha \Rightarrow \lambda_1 = 21\,pm$ Find wavelength of K_β X-ray.
$L_\alpha \Rightarrow \lambda_2 = 140\,pm$

$$E_L - E_K = \frac{hc}{\lambda_1}$$
$$E_M - E_L = \frac{hc}{\lambda_2}$$
$$E_M - E_K = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = \frac{hc}{\lambda}$$

$$\lambda = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2} = \frac{21\times140}{161}\,pm$$

$$\boxed{\lambda = 18.26\,pm}$$

Q Cobalt $\{Z=27\}$, $K_\alpha \Rightarrow \lambda_{Co} = 178.9\,pm$
We found another K_α X-ray $\Rightarrow \lambda_{unknown} = 143.5\,pm$
Identify the impurity in Co.

$$\sqrt{F} = \sqrt{\frac{c}{\lambda}} = a(Z-b)$$

$$\sqrt{\frac{c}{\lambda_{Co}}} = a(27-1)$$

$$\sqrt{\frac{c}{\lambda_{unk}}} = a(Z-1)$$

$$= \sqrt{\frac{178.9}{143.5}} = \frac{Z-1}{26} \Rightarrow Z-1 = 26\sqrt{\frac{178.9}{143.5}}$$

$$Z-1 = 29.03$$

$$\boxed{Z = 30.03}$$ hence, Z=30

∴ Impurity is zinc.

Q Target metal Tungsten $\{Z=74\}$. Find λ of K_β X-rays.

$$\frac{1}{\lambda_{K_\beta}} = R(Z-b)^2\left\{\frac{1}{1^2} - \frac{1}{3^2}\right\}$$

$$= \frac{1}{\lambda_{K_\beta}} = R(74-1)^2\left(\frac{8}{9}\right)$$

$$= \frac{1}{\lambda_{K_\beta}} = R(73)^2\left(\frac{8}{9}\right)$$

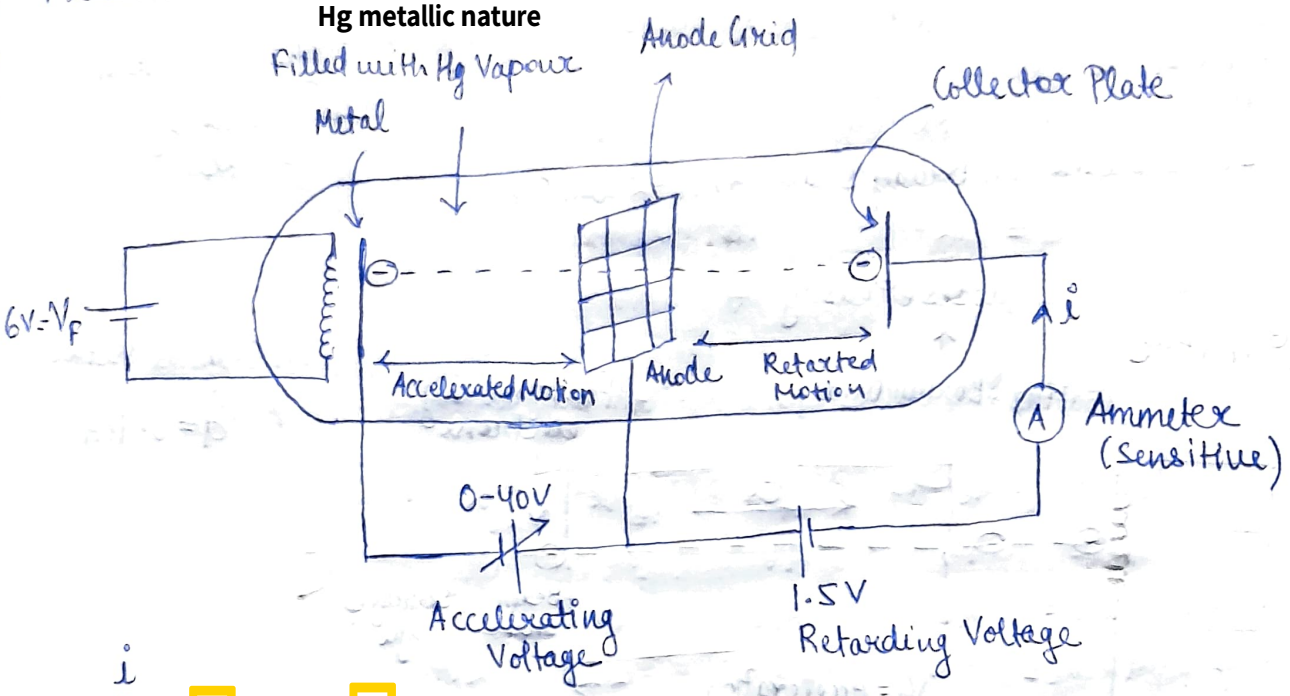
$$\lambda_{K_\beta} = \frac{9}{8\times(73)^2} \times \left(\frac{1}{R}\right)$$

$$\lambda_{K_\beta} = \frac{9}{8\times(73)^2} \times 911\,\text{Å}$$

$$\boxed{\lambda_{K_\beta} = 0.1923\,\text{Å}}$$

Franck-Hertz Experiment : Done by James Franck and Gustav Hertz

Hg metallic nature





* Current drops due to inelastic collision of $e^{\ominus}$ with Hg atom
* The voltage difference b/w any two current drops is always 4.9V as difference in energy level of nth and $(n+1)^{th}$ orbit of Hg atom is 4.9eV.

* Franck-Hertz experiment concludes that atom have specific energy levels only.
* Atoms have specific orbits only called stationary orbits.

Bragg's law :

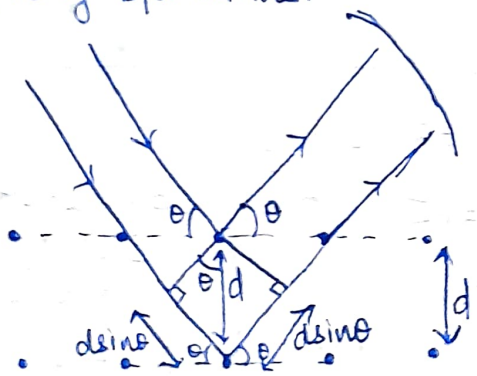
EM waves diffraction slit/hindrance/point/orifice
$a \sim \lambda$

→ EM waves also show diffraction for which slit should be in order of λ.
→ X-ray have λ in order of Å, but for us creating a slit such small is not possible. So we use spacing b/w atoms.

Solid Crystal

d = interatomic spacing

d is of the order of few angstroms



Path difference = Δx

$\Delta x = 2d\sin\theta$

For strongly reflecting/constructive interference :

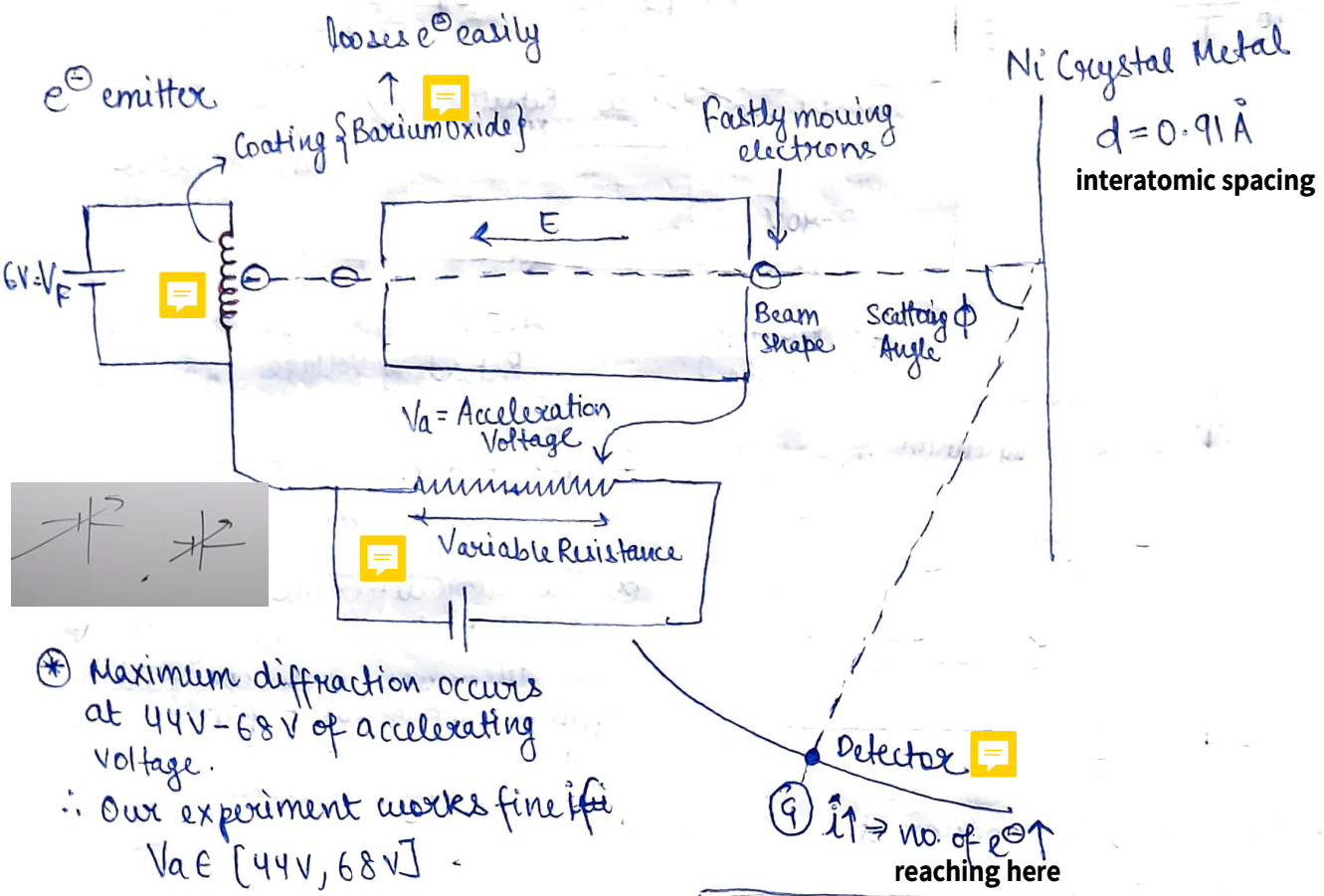
$$\boxed{2d\sin\theta = n\lambda} \quad n \in I^+$$

Bragg's law

→ Used in Crystalography
→ Used to find d.

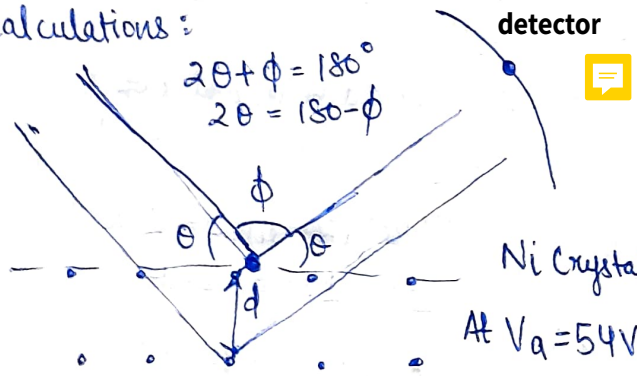
Davison-Germer Experiment (1923-27):

→ $e^{\ominus}$ scattered by the surface of Ni crystal displayed diffraction pattern shown only by waves.
→ Proved de-broglie hypothesis
→ $e^{\ominus}$ can behave like wave
→ It is based on Bragg's law $\{n\lambda = 2d\sin\theta\}$



⊛ Maximum diffraction occurs at 44V - 68V of accelerating voltage.
∴ Our experiment works fine if $V_a \in [44V, 68V]$.

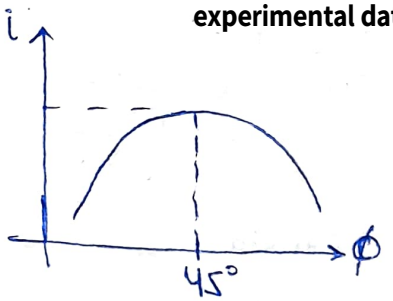
Calculations:



$\Delta x = n\lambda$
$2d\sin\theta = n\lambda$
$2 \times 0.91\text{Å} \times \sin 65° = 1 \times \lambda$
$\lambda = 1.82 \times 0.9063$ Å
$\boxed{\lambda = 1.65\text{Å}}$

At $V_a = 54V \Rightarrow \phi = 50°$
$\therefore \theta = 65°$

Ex If $V_a = 65V$



By de-broglie:

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV_a}}$$

$$\lambda = \sqrt{\frac{150}{V_A}} = \sqrt{\frac{150}{54}} = 1.66\text{Å}$$

⊛ We observe that λ from experiment and de-broglie comes out to be same ⇒ $e^{\ominus}$ behaves as wave.